

SAMIKSHA TRIPATHY

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SUMMARY

Computer Science and Engineering undergraduate specializing in Artificial Intelligence and Machine Learning, with hands-on experience in Python, supervised learning, and backend systems for ML-driven products. Builds and benchmarks models end to end, from data pipeline through model evaluation, and is currently deepening skills in deep learning and computer vision. Seeking a Computer Vision / AI internship.

TECHNICAL SKILLS

Languages Python, Java (Basic), SQL, JavaScript (Basic), HTML, CSS

Machine Learning Scikit-learn, Supervised Learning, Classification, Regression, Random Forest, Model Evaluation and Benchmarking, Neural Network Fundamentals (PyTorch)

Data and Analysis Pandas, NumPy, Matplotlib, Jupyter Notebook, Feature Engineering, Experiment Design

Backend and Tools FastAPI, REST APIs, PostgreSQL, MySQL, SQLAlchemy, Pydantic, Docker, Git, GitHub, pytest, VS Code

AI-Assisted Development Prompt Engineering, Directing AI Coding Tools for Full-Stack Builds, API Integration Design

PROJECTS

AgriOptima AI — Crop Recommendation Engine (Backend Lead) | Python, FastAPI, PostgreSQL, Docker 2026

- Designed and built the entire backend for a crop recommendation platform advising farmers on viable crops from soil, irrigation, season, budget, and location inputs.
- Built the soil and agronomic eligibility engine from scratch, encoding deterministic rules for soil pH, water availability, season, budget, and rotation that reject unsuitable crops with explainable reason codes before scoring.
- Implemented a transparent 0–100 scoring model across soil fit, climate fit, water fit, demand, price, and saturation risk, returning ranked recommendations with score breakdowns and confidence levels.
- Integrated live external data (Open-Meteo weather, data.gov.in AGMARKNET mandi prices) through adapters with timeout handling and labelled cached fallbacks, so no output is presented as more certain than its source.
- Connected all components through a FastAPI service layer with Pydantic validation, SQLAlchemy and Alembic over a Dockerized PostgreSQL database, and an automated pytest suite.

Cache Eviction Policy Benchmark: LRU vs LFU vs Machine Learning | Python, Scikit-learn 2026

- Designed a controlled research benchmark comparing a learned eviction policy against LRU and LFU baselines on a synthetic Zipfian trace with an injected scan to test scan resistance.
- Trained a 200-tree Random Forest regressor on oracle reuse-distance features as an offline approximation of Belady's algorithm, with a strict train/test split to prevent data leakage.
- Measured hit rate and per-eviction latency across a cache-size sweep: the model matched LFU on hit rate (0.559 vs 0.561) while running roughly 14,000x slower per decision, documenting when learned eviction is worth its cost.

Anime Recommendation Platform | AI-Assisted Development, AniList and MyAnimeList APIs 2026

- Directed AI coding tools through iterative prompt engineering to build a full-stack recommendation platform, personally defining features, API integration strategy, and UX flow across two external data sources.

EDUCATION

SRM Institute of Science and Technology Kattankulathur, Tamil Nadu Expected Aug 2028

B.Tech, Computer Science and Engineering — Artificial Intelligence and Machine Learning Specialization | GPA: 9.34 / 10

Relevant Coursework: Machine Learning, Data Structures and Algorithms, DBMS, Operating Systems, Computer Networks

CERTIFICATIONS, LEADERSHIP AND INTERESTS

- Supervised Machine Learning: Regression and Classification — DeepLearning.AI and Stanford Online (Coursera), Andrew Ng.
- Community Volunteer, multiple local NGOs (Ongoing) — contributed time to community-focused initiatives and social outreach.
- Interests: Birding and wildlife photography, digital art and painting, graphic design, UI design.